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Faculty of New Technologies of Information and Communication (NTIC)
Department of Fundamental Computing and its Applications (IFA)

BACHELOR'S THESIS

to obtain the diploma of Bachelor degree in Computer Science

Option: Information Technology (TI)

**Design and implementation of an internship
management and University-Enterprise matching
platform "Stag.io"**

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Dedication

I dedicate this work...

To **my** beloved **Parents**, whose unconditional love, endless sacrifices, and prayers have paved the way for **my** success. You are **my** greatest inspiration, and **I** hope this achievement brings pride to your hearts.

To **my** brothers and sisters, for their constant support and for being **my** lifelong companions in every step of this journey.

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To all those who believe in the power of knowledge and the pursuit of excellence, **I** dedicate this modest fruit of **my** labor.

Abstract

In the rapidly evolving digital landscape, academic institutions face significant challenges in managing the transition of students into the professional world. This thesis presents "**Stag.io**", an innovative web platform **I have designed and implemented** to redefine internship management at the University of Constantine 2.

By implementing a sophisticated "University-Enterprise Matching" mechanism and a centralized administrative dashboard, the system automates complex workflows, from application tracking to agreement validation. Built with a focus on modern UI/UX principles, **Stag.io** not only reduces administrative overhead but also enhances the accessibility and transparency of professional opportunities. The result is a robust, scalable SaaS-style solution that bridges the communication gap between students, supervisors, and industrial partners through a single, intuitive interface.

Keywords: Internship Management, SaaS Platform, Stag.io, University-Enterprise Matching, Workflow Automation.

Résumé

Dans un paysage numérique en constante évolution, les institutions académiques sont confrontées à des défis majeurs pour gérer la transition des étudiants vers le monde professionnel. Cette thèse présente **"Stag.io"**, une plateforme web innovante que **j'ai conçue et implémentée** pour redéfinir la gestion des stages à l'Université de Constantine 2.

En mettant en œuvre un mécanisme sophistiqué de "Matching Université-Entreprise" et un tableau de bord administratif centralisé, le système automatise des flux de travail complexes, allant du suivi des candidatures à la validation des conventions. Conçu avec un accent particulier sur les principes modernes de l'UI/UX, **Stag.io** réduit non seulement la charge administrative, mais améliore également l'accessibilité et la transparence des opportunités professionnelles. Le résultat est une solution robuste et évolutive de type SaaS qui comble le fossé de communication entre les étudiants, les encadrants et les partenaires industriels via une interface unique et intuitive.

Mots-clés : Gestion des stages, Plateforme SaaS, Stag.io, Matching Université-Entreprise, Automatisation des flux de travail.

ملخص

في ظل المشهد الرقمي المتطور بسرعة، تواجه المؤسسات الأكاديمية تحديات كبيرة في إدارة انتقال الطلاب إلى العالم المهني. تقدم هذه الأطروحة منصة "Stag.io"، وهي منصة ويب مبتكرة قمت بتصميمها وتنفيذها لإعادة تعريف إدارة التربصات في جامعة قسنطينة 2.

من خلال تطبيق آلية متطورة لـ "المطابقة بين الجامعة والمؤسسة" ولوحة تحكم إدارية مركزية، يقوم النظام بأتمتة سير العمل المعقد، بدءاً من تتبع الطلبات وصولاً إلى المصادقة على الاتفاقيات. تم بناء Stag.io مع التركيز على مبادئ واجهة وتجربة المستخدم (UI/UX) الحديثة، وهو لا يقلل الأعباء الإدارية فحسب، بل يعزز أيضاً من إمكانية الوصول وشفافية الفرص المهنية. والنتيجة هي حل برمجي قوي وقابل للتوسع بنظام (SaaS) يسد فجوة التواصل بين الطلاب، المشرفين، والشركاء الصناعيين من خلال واجهة واحدة بديهية.

الكلمات المفتاحية: إدارة التربصات، منصة SaaS، Stag.io المطابقة بين الجامعة والمؤسسة، أتمتة سير العمل.

Contents

General Introduction

In the contemporary era, characterized by the rapid acceleration of the Fourth Industrial Revolution, the integration of advanced digital solutions has transcended being a mere administrative luxury to become a fundamental necessity for institutional efficiency. High-performance academic environments, particularly within the Faculty of NTIC at the University of Constantine 2, demand a seamless transition between theoretical knowledge and professional immersion. However, this transition is currently bottlenecked by traditional methodologies that fail to keep pace with the dynamic requirements of the modern Information Technology (IT) sector.

1. Context of the Project

The internship lifecycle represents a critical milestone in the academic journey of a computer science student, acting as the primary bridge to professional life. Within the University of Constantine 2, this process is governed by a complex set of pedagogical and administrative regulations designed to ensure that each student gains relevant industry experience. Despite the global shift towards "Paperless Governance," the local internship management system remains heavily reliant on manual, paper-based workflows. This project, **Stag.io**, emerges as a strategic initiative to align the IFA department's operations with the Algerian Ministry of Higher Education's (MESRS) directives for global digital transformation and the "Digital University" vision.

2. Problem Statement

The reliance on manual workflows introduces a multitude of systemic challenges that hinder the productivity of students, companies, and the university administration. These challenges can be summarized as follows:

- **Informational Blindness:** Students lack a centralized portal to discover opportunities that match their specific technical stacks (e.g., React, Django, or Flutter), leading to mismatched placements.
- **Process Friction:** The physical handling of internship agreements and acceptance letters results in extreme administrative latency, often consuming several weeks of the pedagogical calendar.
- **Data Fragmentation:** There is no unified database to track student progress or company feedback, making it nearly impossible for the administration to conduct

real-time oversight.

3. Contributions

In this work, we propose **Stag.io**, a centralized "Platform-as-a-Service" (PaaS) designed to automate and streamline the entire internship lifecycle. Our platform facilitates a three-way digital ecosystem:

- **For Students:** Providing an intelligent search engine to find internships based on technical "tags" and managing digital profiles.
- **For Recruiters:** Enabling companies to post offers, screen candidates, and manage professional profiles.
- **For Administrators:** Automating the generation of legal internship agreements and providing real-time statistics on student placements.

4. Organization of the Document

This dissertation is organized as follows:

- **Chapter 1 (State of the Art):** Provides a detailed presentation of the project context and a critical analysis of the existing manual system.
- **Chapter 2 (Preliminary Study):** Focuses on identification of actors, functional requirements, and the detailed presentation of the Use Case diagrams.
- **Chapter 3 (Analysis and Design):** Details the system's architecture through UML modeling, including Class and Sequence diagrams.
- **Chapter 4 (Implementation):** Presents the development environment, technologies used (React/Django), and the final results of the system deployment.

Chapter 1

State of the Art

1.1 Introduction

The contemporary era is defined by the rapid acceleration of the Fourth Industrial Revolution, where the transition towards a digital economy is no longer a peripheral institutional goal but a central pillar of modern pedagogical strategy. In this globalized landscape, educational institutions are fundamentally re-evaluating how they bridge the gap between theoretical academic curricula and the highly dynamic demands of professional environments. This bridge is most prominently represented by the professional internship, a critical milestone that serves as the primary incubator for future industry leaders.

However, at the University of Constantine 2, and specifically within the Faculty of NTIC, this crucial transition is currently bottlenecked by legacy methodologies that have failed to keep pace with technological evolution. The persistence of paper-based administrative workflows creates a "digital friction" that not only hinders operational efficiency but also obscures the transparency required for effective pedagogical oversight. The management of hundreds of interns manually leads to fragmented data, informational blindness, and a significant loss of administrative time that could be better spent on student mentorship.

Consequently, this chapter provides a rigorous analytical foundation for the project **Stag.io**. We begin by examining the macro-environmental context of digitalization in higher education, followed by a granular deconstruction of the current ecosystem at the IFA department. By identifying and auditing these systemic inefficiencies—ranging from administrative latency to data vulnerability—we establish a clear and urgent justification for the development of an integrated digital ecosystem. This analytical journey is essential to ensure that **Stag.io** is not merely a digital version of a manual process, but a complete re-engineering of the internship lifecycle to meet international standards of the "Digital

University”.

1.2 Detailed Presentation of the Context

1.2.1 The Paradigm of Digitalization in Academic Environments

The transition towards a digital economy is no longer a peripheral goal but a central pillar of modern pedagogical strategy. In the contemporary academic landscape, information has evolved from a simple resource into the most valuable institutional asset. For institutions like the University of Constantine 2, specifically the Faculty of NTIC, managing student transitions from theoretical modules to professional environments is a critical measure of organizational maturity. Digitalization involves a complete re-engineering of these transitions, ensuring that data flows seamlessly between students, administrative bodies, and industrial partners without the friction of physical barriers or geographical constraints. This shift allows for better resource allocation, real-time tracking of student progress, and a more responsive academic environment that adapts to the rapid fluctuations of the global IT market.

1.2.2 Strategic Alignment with National Directives (MESRS)

The project **Stag.io** is not an isolated initiative; it serves as a technical manifestation of the Algerian Ministry of Higher Education and Scientific Research (MESRS) roadmap for "Global Digital Governance". This national strategy mandates that all pedagogical and administrative interactions be migrated to integrated digital platforms by the end of the decade to ensure efficiency and transparency. By developing a specialized system for internship management, we are moving the IFA department toward a "Zero-Paper" ecosystem. This initiative does not only enhance administrative speed but also ensures strict legal compliance with national standards regarding student data protection, digital signatures, and administrative accountability in the public sector.

1.2.3 Socio-Economic Importance of Professional Internships in IT

In the specific context of Information Technology (IT), internships are the primary vehicle for technology transfer between universities and the local economic fabric. For a student in the third-year license (L3 TI), this period represents the most critical bridge to professional life. A student proficient in advanced frameworks like React for the frontend and Django for the backend requires an environment where these specific skills are leveraged; otherwise, the university's pedagogical investment is significantly undervalued. A robust management system ensures that this human capital is directed towards host companies that maximize the student's technical potential, thereby fostering a culture of

innovation within the national IT industry.

1.3 Comprehensive Presentation of the Existing System (L'Existant)

1.3.1 Operational Deconstruction of the Traditional Manual Workflow

The current manual methodology at the IFA department is fundamentally grounded in a linear and fragile chain of physical document custody. This process can be deconstructed into four critical phases, each presenting its own set of challenges:

- **Exploration and Discovery Phase:** Students independently seek internships through informal networks or physical visits to companies. This creates an "Information Gap," where students with fewer industry connections miss out on high-quality opportunities.
- **The Validation and Acceptance Loop:** Once a company is found, a physical "Lettre d'accueil" is issued. The student must hand-carry this document to the university administration, a step that is highly susceptible to loss, physical damage, or human error during the transcription of data.
- **Manual Pedagogical Review:** A committee manually reviews the internship's subject to ensure it aligns with the L3 TI curriculum. In a department with hundreds of graduating students, this leads to significant administrative bottlenecks and delays in the final academic defense schedule.
- **Clerical Agreement Processing:** The administration manually types the "Convention de Stage". Every clerical error or typo in this legal document requires a full restart of the signature cycle, wasting time, paper, and university resources.

1.3.2 Analytical Audit of Systemic Failure Points

Our analytical audit identifies several "friction points" that make the manual system unsustainable in a modern NTIC faculty:

1. **Administrative Latency:** The average cycle from finding a company to signing the final agreement exceeds 15-20 working days, which often causes students to lose their internship slots.
2. **Informational Blindness:** Administrators have no real-time dashboard to monitor student placements, making it impossible to identify and assist students who are still without an internship late in the semester.

3. **Data Vulnerability and Archiving:** Physical records are vulnerable to environmental damage (fire, humidity) and lack the encryption and automated backup protocols inherent in modern digital databases.

1.4 Similar Applications and Comparative Study

1.4.1 Analysis of Global Professional Platforms (LinkedIn/Indeed)

Generic platforms offer global networking but lack "Institutional Intelligence". They cannot generate the official "Convention de Stage" required by the Algerian MESRS, nor do they support the specific internal validation workflow required by the IFA department.

1.4.2 Strategic Comparison and Synthesis Table

The following table provides a comparison of internship management approaches:

Table 1.1: Strategic Comparison of Internship Management Solutions			
Criterion	Manual (Current)	LinkedIn	Stag.io
Institutional Compliance	High (but slow)	None	High (Automated)
Operational Speed	Very Slow (Weeks)	Medium	Real-time (Minutes)
Analytical Reporting	Non-existent	Limited	Advanced Dashboards
Data Integrity	Vulnerable (Paper)	Digital	Encrypted/Secure

1.5 Synthesis and Strategic Justification

The comprehensive analysis of the "State of the Art" conducted in this chapter provides irrefutable evidence that the current "Paper-First" approach at the University of Constantine 2 is no longer sustainable. This methodology acts as a severe structural bottleneck, creating a cascading series of inefficiencies that affect all stakeholders involved in the internship lifecycle.

1.5.1 Synthesizing the Critical Failures

The existing system’s failure is not merely administrative but systemic. The synthesis of our study highlights three core dimensions of this crisis:

- **The Operational Dimension:** The manual workflow is characterized by high latency and clerical redundancy. The time spent on physical document transit and manual data entry represents a significant waste of human capital, both for students and administrative staff.
- **The Pedagogical Dimension:** Due to the lack of a centralized digital portal, there is a "Skill Mismatch" between the technical profiles of Information Technology (TI) students and the actual requirements of host enterprises. The manual system lacks the analytical depth to match skills with opportunities effectively.
- **The Institutional Dimension:** The absence of real-time monitoring and digital archiving leaves the administration in a state of "informational blindness," making it impossible to perform long-term statistical analysis or align with the national "Zero-Paper" strategy (MESRS).

1.5.2 Stag.io: The Evolutionary Necessity

In light of these findings, the development of **Stag.io** is not proposed as a mere digital alternative or a secondary tool; it is presented as a **necessary technical evolution**. Our solution is designed to act as a centralized "Digital Bridge" that re-engineers the entire Verification-Validation-Certification cycle.

By integrating modern web architectures—specifically the React-Django technical stack—**Stag.io** provides the scalability and data integrity required to align the IFA department with international digital standards. The platform will effectively transition the department from a reactive, paper-dependent entity to a proactive, data-driven ecosystem.

1.5.3 Transition to Preliminary Study

The conclusion of this analytical phase confirms that the technical mandate for **Stag.io** is clearly defined. Having established the "Why" (the justification) and the "What" (the context) in this chapter, it becomes imperative to move towards the "How." The following chapter will focus on the **Preliminary Study**, where we will deconstruct the functional requirements, identify the key actors, and define the technical specifications necessary to bring this digital ecosystem to fruition.

Chapter 2

Preliminary Study and Needs Specification

2.1 Introduction

This chapter represents the analytical cornerstone of the **Stag.io** project, serving as the formal transition from conceptual ambition to structural realization. It serves to bridge the critical gap between the initial socio-pedagogical problem statement and the rigorous technical design phase that follows. In the context of the global digital transformation of higher education, the management of professional internships has emerged as a complex administrative challenge that requires a standardized, scalable solution capable of operating across diverse academic environments.

While the project originates from a specific academic context, its architectural philosophy is designed to be **department-agnostic** and **universally adaptable**. **Stag.io** is conceived not merely as a local tool, but as a modular ecosystem that addresses the universal requirements of the internship lifecycle—from the initial student-company matching to the complex legal generation of administrative agreements—regardless of the specific faculty or institutional discipline.

By meticulously identifying the universal actors involved in this process and deconstructing the multi-layered functional requirements, we establish a comprehensive blueprint. This ensures that the final system aligns with the strategic pedagogical goals and administrative governance standards mandated by national educational directives. This chapter, therefore, provides the functional justification for a platform that aims to unify the internship experience for students, recruiters, and university administrations on a macro scale.

2.2 Presentation of the Project to be Realized

The project **Stag.io** is conceived as a sophisticated, integrated web-based ecosystem engineered to fully digitalize and optimize the professional internship lifecycle. Its fundamental mission is to replace obsolete, legacy paper-based workflows with a high-performance dynamic platform where all stakeholders—students, industrial partners, and university administrations—can interact within a transparent, secure, and highly efficient digital environment.

To achieve this ambitious transformation, the system is architected around three core strategic pillars:

2.2.1 Automation of Regulatory and Legal Documentation

The primary administrative bottleneck in the current system is the manual generation and physical transit of legal documents. **Stag.io** introduces an automated document engine that dynamically populates the official "Convention de Stage" and "Lettre d'accueil" using verified data from the system's central database. By leveraging pre-defined templates that comply with national administrative standards (MESRS), the system eliminates clerical errors, reduces the document processing cycle from weeks to minutes, and ensures that all parties have immediate access to legally sound documentation.

2.2.2 Real-time Analytical Tracking of Placements

Transparency is a key deficiency in traditional manual systems. **Stag.io** provides university administrators with advanced real-time dashboards to monitor the entire placement landscape. This pillar enables the tracking of student progress—from the initial search phase to the final validation—allowing institutions to proactively identify students who may be struggling to secure a placement. Furthermore, this continuous data flow facilitates the generation of academic statistics and placement reports, which are essential for long-term institutional strategic planning.

2.2.3 Skill-Based Intelligent Matching Engine

Beyond simple administrative management, **Stag.io** acts as a professional bridge between academic curricula and industry needs. Through a technical matching engine, the system analyzes student technical profiles—specifically focusing on technologies like React, Django, and Flutter—and aligns them with the specific "Skill Tags" requested by recruiters in their offers. This ensures that students are not merely placed in any company, but are matched with professional environments that maximize their specific technical potential and foster genuine technology transfer between the university and the

industrial sector.

2.3 Identification of the Actors

In accordance with the system's operational scope and the functional boundaries defined in the UML modeling phase, three primary actors have been identified. Each actor represents a specific entity that interacts with the "Stag.io" system to fulfill a set of strategic objectives within the internship lifecycle.

2.3.1 The Student (The Core Beneficiary)

The student is the primary end-user and the central beneficiary of the platform's ecosystem. Beyond simple browsing, the student acts as a "Digital Professional" within the system. Their responsibilities and interactions include:

- **Identity and Profile Engineering:** The student is responsible for the "Manage CV" use case, where they must structure their technical competencies into searchable metadata.
- **Proactive Opportunity Discovery:** Utilizing the "Search Internships" function, the student filters through institutional offers to find placements that align with their career goals.
- **Transaction Tracking:** Once an application is submitted through the "Apply for Internship" use case, the student must monitor the real-time feedback loop provided by the system to track their selection status.

2.3.2 The Company Recruiter (The Industrial Partner)

The Recruiter represents the external organizational entity seeking technical talent. This actor bridges the gap between industrial needs and academic potential through the following actions:

- **Strategic Offer Management:** Through the "Manage Offers" use case, the recruiter defines the technical requirements, duration, and pedagogical goals of the internship slots available.
- **Candidate Evaluation and Selection:** The recruiter performs the "View Candidates" action to access student profiles. They have the final decision-making power in the "Accept / Reject Candidate" use case, which triggers the next phase of the administrative workflow.

- **Institutional Collaboration:** By managing their company profile, the recruiter establishes a long-term digital presence within the university’s partner network.

2.3.3 The University Administrator (The Regulatory Authority)

The Admin serves as the system’s moderator and the legal guarantor of the internship’s academic validity. This role is vital for ensuring that the transition from university to enterprise follows institutional standards:

- **Pedagogical Oversight:** The Admin executes the ”Validate Internship” use case, reviewing the internship’s subject to ensure it meets the required scientific and technical level of the student’s curriculum.
- **Operational Governance:** Through ”Track Placements” and ”View Statistics,” the Admin maintains a macro-view of the entire placement process, ensuring no student is left behind.
- **Legal Document Execution:** The Admin is the sole actor responsible for triggering the ”Generate Agreement” use case, which transforms a validated placement into a legally binding ”Convention de Stage”.

2.4 Cahier des Charges (Requirements Specification)

The specification of requirements is a fundamental step that defines the functional and technical boundaries of **Stag.io**. It serves as a formal contract between the conceptual design and the final implementation, ensuring all stakeholder needs are met with precision.

2.4.1 Functional Requirements (Besoins Fonctionnels)

Functional requirements describe the specific services and behaviors the system must exhibit. For **Stag.io**, these are categorized into high-priority modules:

Identity and Access Governance

- **Secure Multi-Actor Authentication:** The system must implement a robust authentication layer where students, recruiters, and administrators can access their respective dashboards using encrypted credentials.
- **Role-Based Access Control (RBAC):** Beyond simple login, the system must strictly enforce permissions. For instance, a student cannot access the administrative validation panel, and a recruiter cannot modify university-level templates.

Profile and Skill Engineering Module

- **Dynamic Digital CV Builder:** Students must be provided with tools to construct a structured professional profile. This includes the ability to categorize technical skills (e.g., React, Django) into tags that the system’s matching engine can process.
- **Corporate Identity Management:** Recruiters must be able to manage a comprehensive company profile, including legal descriptions and past internship history, to establish institutional trust.

Internship Marketplace and Lifecycle Management

- **Advanced Matching and Filtering:** The system must allow students to filter offers not just by date, but by ”Skill-Match” scores, identifying opportunities that perfectly align with their L3 TI specializations.
- **Real-time Application Workflow:** When a student applies, the system must trigger a transactional workflow that notifies the recruiter and updates the student’s ”Tracking Dashboard” in real-time.

Administrative and Legal Automation

- **Automated Validation Cycle:** The system must provide the Administrator with a dedicated interface to review and ”Validate” or ”Reject” student-company matches based on pedagogical criteria.
- **Document Generation Engine:** Upon validation, the system must automatically execute a PDF generation script that pulls metadata (Student ID, Company Tax ID, Dates) into the official university ”Convention de Stage” template, eliminating all manual typing errors.

2.4.2 Non-Functional Requirements (Besoins Non-Fonctionnels)

These requirements define the quality attributes, ensuring the system is professional, reliable, and scalable:

System Performance and Scalability

The platform must maintain a high Quality of Service (QoS). Specifically, the system must handle the concurrent load of hundreds of students during the peak internship search month (typically March to May) without exceeding a response time of 2 seconds for critical queries.

Security and Data Sovereignty

Given that the system handles sensitive personal data and institutional agreements, security is paramount:

- **Encryption:** All passwords must be hashed using industry-standard algorithms (e.g., Argon2 or BCrypt).
- **Data Integrity:** The PostgreSQL database must implement referential integrity to ensure that an internship agreement cannot exist without a validated student and company.

Technical Usability and Interoperability

The user interface must adhere to the "Mobile-First" philosophy, ensuring that students can track their applications via smartphones as easily as on a desktop. Furthermore, the system architecture should be designed to allow future integration with larger university management systems (e.g., Progress).